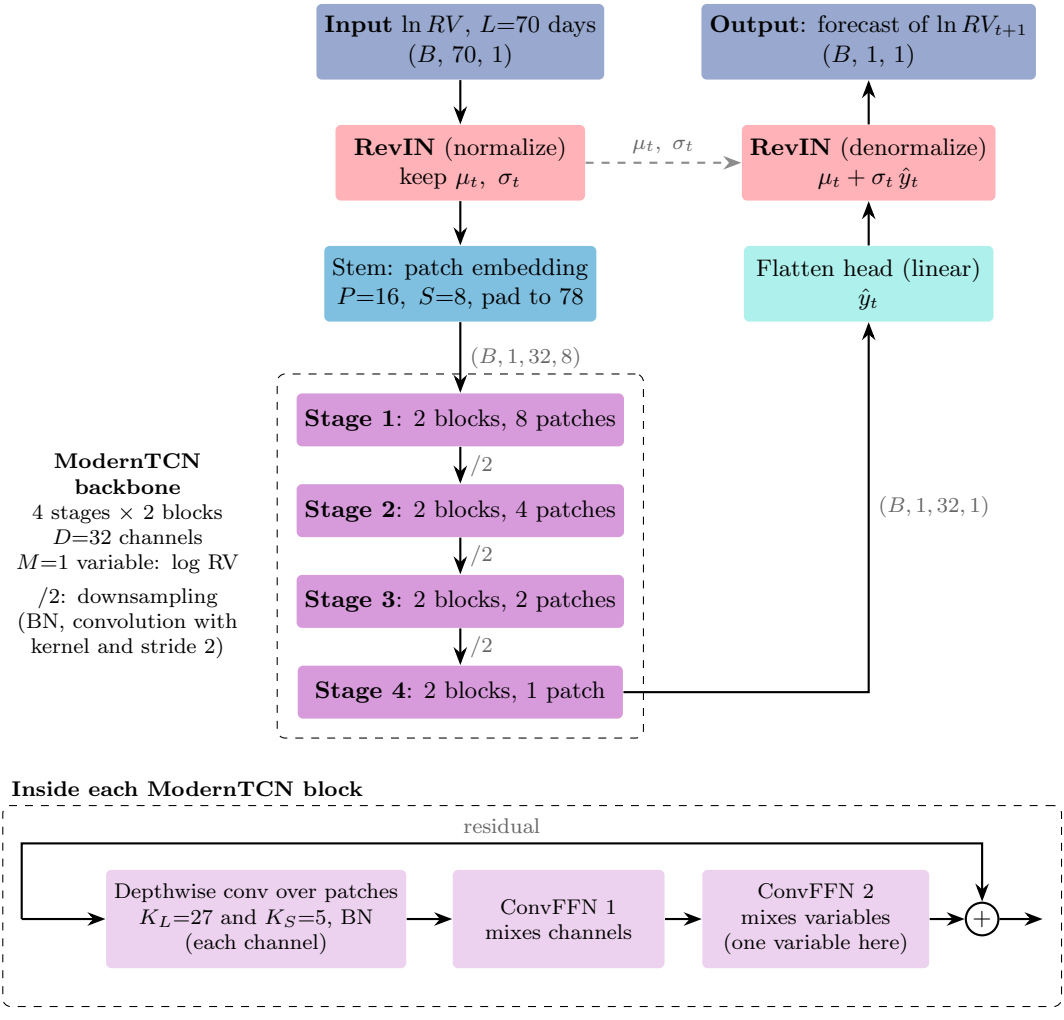

Input ln RV, L=70 days
$(B, 70, 1)$
Output: forecast of ln RV_{t+1}
$(B, 1, 1)$
RevIN (normalize)
keep μ_t, σ_t
μ_t, σ_t
RevIN (denormalize)
$\mu_t + \sigma_t \hat{y}_t$
Stem: patch embedding
P=16, S=8, pad to 78
Flatten head (linear)
$\hat{y}_t$
$(B, 1, 32, 8)$
$(B, 1, 32, 1)$
Stage 1: 2 blocks, 8 patches
/2
Stage 2: 2 blocks, 4 patches
/2
Stage 3: 2 blocks, 2 patches
/2
Stage 4: 2 blocks, 1 patch
ModernTCN backbone
4 stages × 2 blocks
D=32 channels
M=1 variable: log RV
/2: downsampling (BN, convolution with kernel and stride 2)
Inside each ModernTCN block
residual
Depthwise conv over patches
K_L=27 and K_S=5, BN
(each channel)
ConvFFN 1
mixes channels
ConvFFN 2
mixes variables
(one variable here)